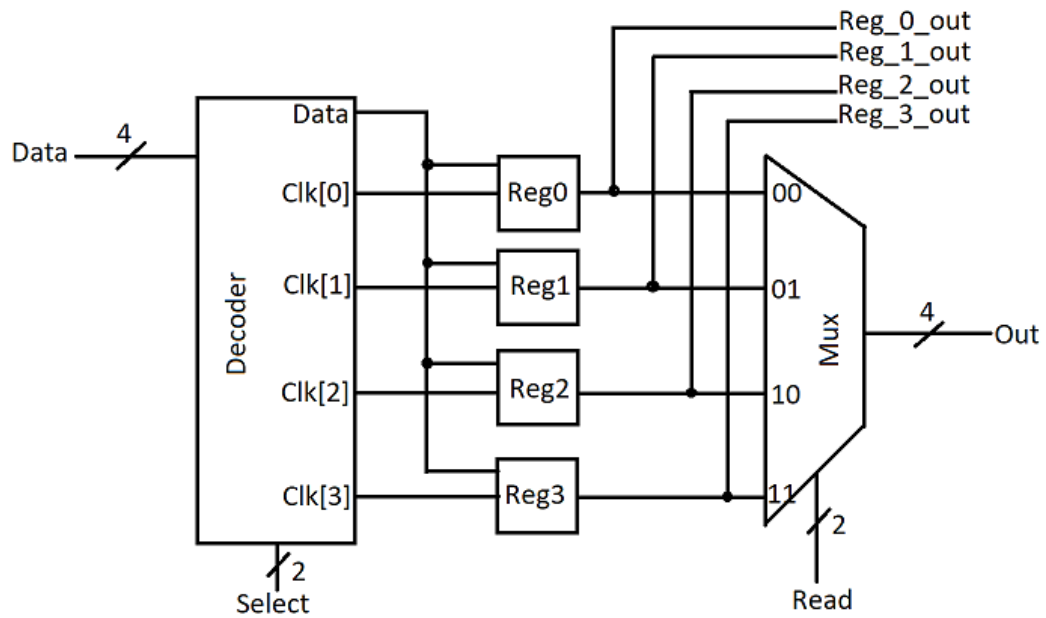


Practice #5: Basic sequential circuits with Verilog in Cadence

Make 4 registers of 4 bits each, that you can select where you'll send the data so the register can save the value using the Clk of the register. And in the Output use a multiplexer to choose which output you want to see

HDL codes



```

1 //Verilog HDL for "Verilog", "Multiplexed_Register" "functional"
2
3
4 module Multiplexed_Register (Data,Select,Read,Out,Reg_0_out,Reg_1_out,Reg_2_out,Reg_3_out);
5
6     input  [3:0] Data;
7     input  [1:0] Select;
8     input  [1:0] Read;
9     output [3:0] Out, Reg_0_out, Reg_1_out, Reg_2_out, Reg_3_out;
10    reg    [3:0] Clk;
11    reg    [3:0] Reg_0, Reg_1, Reg_2, Reg_3;
12
13    always @(*) begin
14        case(Select)
15            2'b00: Clk = 4'b0001;
16            2'b01: Clk = 4'b0010;
17            2'b10: Clk = 4'b0100;
18            2'b11: Clk = 4'b1000;
19            default: Clk = 4'bxxxx;
20        endcase
21    end
22
23    always @(posedge Clk[0]) begin
24        Reg_0 <= Data;
25    end
26
27    always @(posedge Clk[1]) begin
28        Reg_1 <= Data;
29    end
30
31    always @(posedge Clk[2]) begin
32        Reg_2 <= Data;
33    end
34
35    always @(posedge Clk[3]) begin
36        Reg_3 <= Data;
37    end
38
39    assign Out = Read[1] ? (Read[0] ? Reg_3 : Reg_2)
40                : (Read[0] ? Reg_1 : Reg_0);
41
42    assign Reg_0_out = Reg_0;
43    assign Reg_1_out = Reg_1;
44    assign Reg_2_out = Reg_2;
45    assign Reg_3_out = Reg_3;
46
47 endmodule

```

```

Registers.v
1  module register (Data,Select,Read,Out,Clk);
2
3      input  [3:0] Data;
4      input  [1:0] Select;
5      input  [1:0] Read;
6      output [3:0] Out;
7      input Clk;
8      reg    [3:0] Reg_0, Reg_1, Reg_2, Reg_3;
9
10     always @(posedge Clk) begin
11         if (Select == 2'b00) begin
12             Reg_0 <= Data;
13         end
14         else if (Select == 2'b01) begin
15             Reg_1 <= Data;
16         end
17         else if (Select == 2'b10) begin
18             Reg_2 <= Data;
19         end
20         else if (Select == 2'b11) begin
21             Reg_3 <= Data;
22         end
23     end
24
25     assign Out = Read[1] ? (Read[0] ? Reg_3 : Reg_2)
26                 : (Read[0] ? Reg_1 : Reg_0);
27
28
29     endmodule

```

Test bench codes

```

1 //Verilog HDL for "Verilog", "Multiplexed_Register_tb" "functional"
2
3
4 module Multiplexed_Register_tb ( );
5
6 reg [3:0] Data;
7 reg [1:0] Select;
8 reg [1:0] Read;
9 wire [3:0] Out, Reg_0_out, Reg_1_out, Reg_2_out, Reg_3_out;
10 wire [3:0] Clk;
11 wire [3:0] Reg_0, Reg_1, Reg_2, Reg_3;
12
13 Multiplexed_Register uut(Data,Select,Read,Out,Reg_0_out,Reg_1_out,Reg_2_out,Reg_3_out);
14 initial begin
15
16     $display("Start");
17
18     Data = 4'b1101; //D
19     Select = 2'b00;
20     Read = 2'b00;
21     #20;
22     $display("A");
23     Data = 4'b1111; //F
24     Select = 2'b01;
25     Read = 2'b01;
26     #20;
27     $display("B");
28     Data = 4'b0110; //6
29     Select = 2'b10;
30     Read = 2'b10;
31     #20;
32     $display("C");
33     Data = 4'b1000; //8
34     Select = 2'b11;
35     Read = 2'b11;
36     #20;
37     $display("D");
38
39     Data = 4'b1010; //A
40     Select = 2'b00;
41     Read = 2'b00;
42     #20;
43     $display("E");
44     Data = 4'b0001; //1
45     Select = 2'b01;
46     Read = 2'b01;
47     #20;
48
49     $display("F");
50     Data = 4'b0111; //7
51     Select = 2'b10;
52     Read = 2'b10;
53     #20;
54     $display("G");
55     Data = 4'b0011; //3
56     Select = 2'b11;
57     Read = 2'b11;
58     #20;
59     $display("H");
60     $display("Test completed");
61
62 end
63
64 endmodule
65

```

```

Register_tb.v
1  //MULTIPLEXED REGISTER TEST BENCH
2  `timescale 1ns/1ns
3  `include "Registers.v" //Importamos el archivo al cual le hacemos su testbench
4
5  module register_tb ();
6
7  reg [3:0] Data;
8  reg [1:0] Select;
9  reg [1:0] Read;
10 wire [3:0] Out;
11 reg Clk;
12 wire [3:0] Reg_0, Reg_1, Reg_2, Reg_3;
13
14 register uut(Data,Select,Read,Out,Clk);
15
16 initial begin
17
18     $dumpfile("Register_tb.vcd");
19     $dumpvars(0,register_tb);
20     $display("Start");
21
22     Clk = 0;
23     #10;
24     Clk = 1;
25     #10;
26     Clk = 0;
27     #10;
28
29     Data = 4'b1101; //D
30     Select = 2'b00;
31     Read = 2'b00;

```

```

31     Read = 2'b00;
32
33     Clk = 1;
34     #10;
35     Clk = 0;
36     #10;
37
38     Data = 4'b0011; //D
39
40     #10;
41     Clk = 0;
42     #10;
43     Clk = 1;
44     #10;
45     Clk = 0;
46     #10;
47
48     $display("A");
49     Data = 4'b1111; //F
50     Select = 2'b01;
51     Read = 2'b01;
52     #20;
53
54     Clk = 0;
55     #10;
56     Clk = 1;
57     #10;
58     Clk = 0;
59     #10;

```

```
61    $display("B");
62    Data = 4'b0110; //6
63    Select = 2'b10;
64    Read = 2'b10;
65    #20;
66
67    Clk = 0;
68    #10;
69    Clk = 1;
70    #10;
71    Clk = 0;
72    #10;
73
74    $display("C");
75    Data = 4'b1000; //8
76    Select = 2'b11;
77    Read = 2'b11;
78    #20;
79
80    Clk = 0;
81    #10;
82    Clk = 1;
83    #10;
84    Clk = 0;
85    #10;
86
87    $display("D");
88
89    Data = 4'b1010; //A
```

```

90     Select = 2'b00;
91     Read = 2'b00;
92     #20;
93
94     Clk = 0;
95     #10;
96     Clk = 1;
97     #10;
98     Clk = 0;
99     #10;
100
101     $display("E");
102     Data = 4'b0001; //1
103     Select = 2'b01;
104     Read = 2'b01;
105     #20;
106
107     Clk = 0;
108     #10;
109     Clk = 1;
110     #10;
111     Clk = 0;
112     #10;
113
114     $display("F");
115     Data = 4'b0111; //7
116     Select = 2'b10;
117     Read = 2'b10;
118     #20;

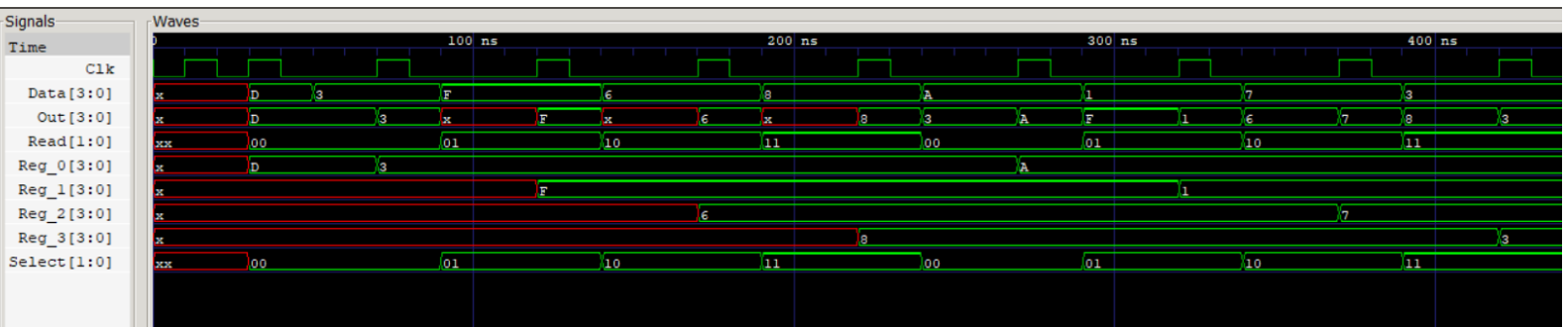
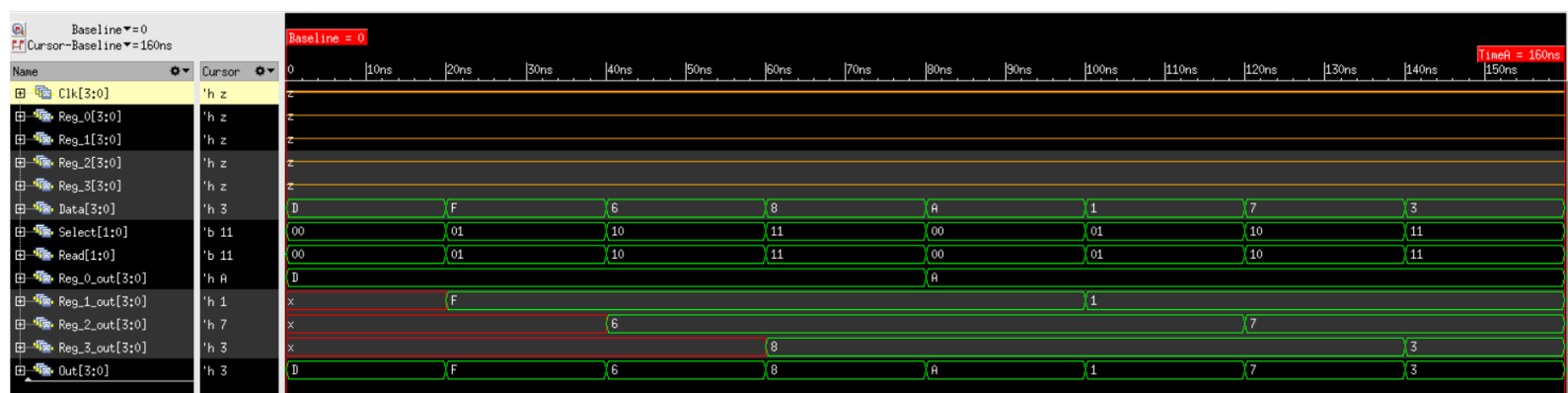
```

```

120         Clk = 0;
121         #10;
122         Clk = 1;
123         #10;
124         Clk = 0;
125         #10;
126
127         $display("G");
128         Data = 4'b0011; //3
129         Select = 2'b11;
130         Read = 2'b11;
131         #20;
132         Clk = 0;
133         #10;
134         Clk = 1;
135         #10;
136         Clk = 0;
137         #10;
138         $display("H");
139
140         $display("Test completed");
141
142     end
143
144 endmodule
145

```

Simulation



Codes

//Verilog HDL for "Verilog", "Multiplexed_Register" "functional"

```
module Multiplexed_Register
(Data,Select,Read,Out,Reg_0_out,Reg_1_out,Reg_2_out,Reg_3_out);
```

```
    input [3:0] Data;
    input [1:0] Select;
    input [1:0] Read;
    output [3:0] Out, Reg_0_out, Reg_1_out, Reg_2_out, Reg_3_out;
    reg [3:0] Clk;
    reg [3:0] Reg_0, Reg_1, Reg_2, Reg_3;
```

```
    always @(*) begin
```

```
        case(Select)
```

```
            2'b00: Clk = 4'b0001;
```

```
            2'b01: Clk = 4'b0010;
```

```
            2'b10: Clk = 4'b0100;
```

```
            2'b11: Clk = 4'b1000;
```



```

        default: Clk = 4'bxxxx;
    endcase
end

always @(posedge Clk[0]) begin
    Reg_0 <= Data;
end

always @(posedge Clk[1]) begin
    Reg_1 <= Data;
end

always @(posedge Clk[2]) begin
    Reg_2 <= Data;
end

always @(posedge Clk[3]) begin
    Reg_3 <= Data;
end

assign Out = Read[1] ? (Read[0] ? Reg_3 : Reg_2)
              : (Read[0] ? Reg_1 : Reg_0);

assign Reg_0_out = Reg_0;
assign Reg_1_out = Reg_1;
assign Reg_2_out = Reg_2;
assign Reg_3_out = Reg_3;

endmodule

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

//Verilog HDL for "Verilog", "Multiplexed_Register_tb" "functional"

module Multiplexed_Register_tb ( );

reg [3:0] Data;
reg [1:0] Select;
reg [1:0] Read;
wire [3:0] Out, Reg_0_out, Reg_1_out, Reg_2_out, Reg_3_out;
wire [3:0] Clk;
wire [3:0] Reg_0, Reg_1, Reg_2, Reg_3;

Multiplexed_Register uut(Data,Select,Read,Out,Reg_0_out,Reg_1_out,Reg_2_out,Reg_3_out);

```

initial begin

```
$display("Start");
```

```
Data = 4'b1101; //D
```

```
Select = 2'b00;
```

```
Read = 2'b00;
```

```
#20;
```

```
$display("A");
```

```
Data = 4'b1111; //F
```

```
Select = 2'b01;
```

```
Read = 2'b01;
```

```
#20;
```

```
$display("B");
```

```
Data = 4'b0110; //6
```

```
Select = 2'b10;
```

```
Read = 2'b10;
```

```
#20;
```

```
$display("C");
```

```
Data = 4'b1000; //8
```

```
Select = 2'b11;
```

```
Read = 2'b11;
```

```
#20;
```

```
$display("D");
```

```
Data = 4'b1010; //A
```

```
Select = 2'b00;
```

```
Read = 2'b00;
```

```
#20;
```

```
$display("E");
```

```
Data = 4'b0001; //1
```

```
Select = 2'b01;
```

```
Read = 2'b01;
```

```
#20;
```

```
$display("F");
```

```
Data = 4'b0111; //7
```

```
Select = 2'b10;
```

```
Read = 2'b10;
```

```
#20;
```

```
$display("G");
```

```
Data = 4'b0011; //3
```

```
Select = 2'b11;
```

```
Read = 2'b11;
```

```
#20;
```

```

    $display("H");

    $display("Test completed");

end

endmodule
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
module register (Data,Select,Read,Out,Clk);

    input [3:0] Data;
    input [1:0] Select;
    input [1:0] Read;
    output [3:0] Out;
    input Clk;
    reg [3:0] Reg_0, Reg_1, Reg_2, Reg_3;

    always @(posedge Clk) begin
        if (Select == 2'b00) begin
            Reg_0 <= Data;
        end
        else if (Select == 2'b01) begin
            Reg_1 <= Data;
        end
        else if (Select == 2'b10) begin
            Reg_2 <= Data;
        end
        else if (Select == 2'b11) begin
            Reg_3 <= Data;
        end
    end

    assign Out = Read[1] ? (Read[0] ? Reg_3 : Reg_2)
                : (Read[0] ? Reg_1 : Reg_0);

endmodule
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
//MULTIPLEXED REGISTER TEST BENCH
`timescale 1ns/1ns
`include "Registers.v" //Importamos el archivo al cual le hacemos su testbench

module register_tb ();

```

```
reg [3:0] Data;
reg [1:0] Select;
reg [1:0] Read;
wire [3:0] Out;
reg Clk;
wire [3:0] Reg_0, Reg_1, Reg_2, Reg_3;

register uut(Data,Select,Read,Out,Clk);
```

```
initial begin
```

```
    $dumpfile("Register_tb.vcd");
    $dumpvars(0,register_tb);
    $display("Start");
```

```
    Clk = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
```

```
    Data = 4'b1101; //D
    Select = 2'b00;
    Read = 2'b00;
```

```
    Clk = 1;
    #10;
    Clk = 0;
    #10;
```

```
    Data = 4'b0011; //D
```

```
    #10;
    Clk = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
```

```
    $display("A");
    Data = 4'b1111; //F
    Select = 2'b01;
```

```
Read = 2'b01;  
#20;
```

```
Clk = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;
```

```
$display("B");  
Data = 4'b0110; //6  
Select = 2'b10;  
Read = 2'b10;  
#20;
```

```
Clk = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;
```

```
$display("C");  
Data = 4'b1000; //8  
Select = 2'b11;  
Read = 2'b11;  
#20;
```

```
Clk = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;
```

```
$display("D");
```

```
Data = 4'b1010; //A  
Select = 2'b00;  
Read = 2'b00;  
#20;
```

```
Clk = 0;
```

```
#10;
Clk = 1;
#10;
Clk = 0;
#10;

$display("E");
Data = 4'b0001; //1
Select = 2'b01;
Read = 2'b01;
#20;

    Clk = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;

$display("F");
Data = 4'b0111; //7
Select = 2'b10;
Read = 2'b10;
#20;

    Clk = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;

$display("G");
Data = 4'b0011; //3
Select = 2'b11;
Read = 2'b11;
#20;
    Clk = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;
$display("H");
```

```
$display("Test completed");
```

```
end
```

```
endmodule
```